

HARMONICDIFF

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Problem

Let m, n be positive integers greater than 1. We define the sets

$P_m = \left\{ \frac{1}{m}, \frac{2}{m}, \dots, \frac{m-1}{m} \right\}$ and $P_n = \left\{ \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n} \right\}$. Find the distance between P_m and P_n , that is defined as

$$\min\{|a - b| : a \in P_m, b \in P_n\}.$$

Solution. Let the answer be $D(m, n)$. We claim that:

$$D(m, n) = \begin{cases} 0, & \text{if } \gcd(m, n) \geq 1 \\ \frac{1}{\text{lcm}(m, n)}, & \text{if } \gcd(m, n) = 1 \end{cases}$$

We will prove our claim with appropriate casework:

- $\gcd(m, n) = d \geq 1$: Then, let $a = \frac{m}{d}$ and $b = \frac{n}{d}$. Observe that $\left| \frac{a}{m} - \frac{b}{n} \right| = 0$. Since $|x| \geq 0$ for all x , 0 is the minimum possible value.

- $\gcd(m, n) = 1$: First of all, observe that this cannot be zero because if $\exists a_0, b_0$ such that $\left| \frac{a_0}{m} - \frac{b_0}{n} \right| = 0$ then we would have $\frac{m}{n} = \frac{a_0}{b_0}$ but since m, n are co-prime, so $a_0 \geq m, b_0 \geq n$. Since we have $a_0 \leq m-1, b_0 \leq n-1$, this is impossible.

Due to Bezout's Lemma, $\exists a, b \in \mathbb{Z}$ such that $ma + nb = 1$ ((wolog $a < 0$) and $\left\lfloor \frac{a}{n} \right\rfloor = \left\lfloor \frac{-b}{m} \right\rfloor$. Reducing $a \pmod n, b \pmod m$ such that $\left\lfloor \frac{a}{n} \right\rfloor = 0$, for any x, y

we have $\left| \frac{x}{m} - \frac{y}{n} \right| = \left| \frac{mx - ny}{mn} \right| \geq \frac{1}{mn}$. Taking $x = a, y = -b$, we get that the lower bound is attainable.